

1 Ideal Over-the-Air Computation Baseline

Before introducing the practical impairments inherent to physical wireless channels—such as multipath delay spread, intersymbol interference, carrier frequency offsets, and time synchronization errors—it is essential to establish an analytical and simulation baseline of Over-the-Air (OTA) computation under ideal conditions. The foundational premise of OTA computing is its ability to overcome the communication bottleneck of distributed wireless sensor aggregation. In conventional orthogonal multi-access protocols (e.g., Time-Division Multiple Access (TDMA) or Orthogonal Frequency-Division Multiplexing (OFDM)), the time required to collect readings from K distributed End Device (ED)s scales linearly with the network size, incurring a latency complexity of $\mathcal{O}(K)$. By contrast, OTA computing exploits the superposition property of the wireless Multiple Access Channel (MAC) to execute nomographic function aggregation directly across the physical propagation medium in a single time slot:

$$\mathcal{O}(K) \longrightarrow \mathcal{O}(1) \quad (1)$$

In this idealized baseline scenario, all transmitting nodes are assumed to be perfectly synchronized in carrier frequency, phase, and symbol timing. The propagation channel is modeled as a non-dispersive, flat-fading medium corrupted solely by Additive White Gaussian Noise (AWGN). This chapter presents an end-to-end multi-tiered experimental workflow that demonstrates this unimpaired baseline:

- (i) The physical principles and electrical circuit equivalent of electromagnetic superposition;
- (ii) The continuous-time and passband Amplitude Modulation (AM) mathematical formulation;
- (iii) Algorithmic discrete-time MATLAB modeling and multi-rate pulse-shaping code;
- (iv) Mean squared error and constellation diagnostic evaluations;
- (v) Analog eye diagram analysis comparing MATLAB and Simulink dynamics;
- (vi) Multi-rate block-level Simulink dynamical modeling; and
- (vii) Software-Defined Radio (Software Defined Radio (SDR)) realization and polyphase filter calibration in GNU Radio.

Establishing this rigorous benchmark provides the mathematical reference and experimental framework repeated throughout subsequent chapters to evaluate model impurities and channel imperfections.

1.a Physical Principles and Electrical Circuit Equivalent of Electromagnetic Superposition

In conventional digital signal processors, arithmetic computation is executed through Boolean logic gates constructed from semiconductor transistors. In OTA computation, however, arithmetic addition is performed natively by free space through Maxwell's electrodynamic equations.

When K distributed sensor nodes concurrently radiate radio-frequency currents through their respective antennas, the emitted electric field vectors $\mathbf{E}_k(\mathbf{r}, t)$ superimpose linearly in the spatial continuum according to the principle of electromagnetic superposition:

$$\mathbf{E}_{\text{total}}(\mathbf{r}, t) = \sum_{k=1}^K \mathbf{E}_k(\mathbf{r}, t) \quad (2)$$

At the receive antenna aperture of the central Fusion Center (FC), the incident electric field induces an open-circuit terminal voltage proportional to the spatial integral of the field weighted by the antenna's effective vector length.

This electrodynamic mechanism is formalized electrically in Figure 1. Each sensor node is modeled as a Thévenin equivalent source generating an information-bearing voltage $v_k(t)$ with internal impedance R_{S_k} . The radiating element is characterized by an equivalent resonant RLC circuit with antenna radiation resistance $R_{\text{rad},k}$. The wireless medium behaves as an unguided

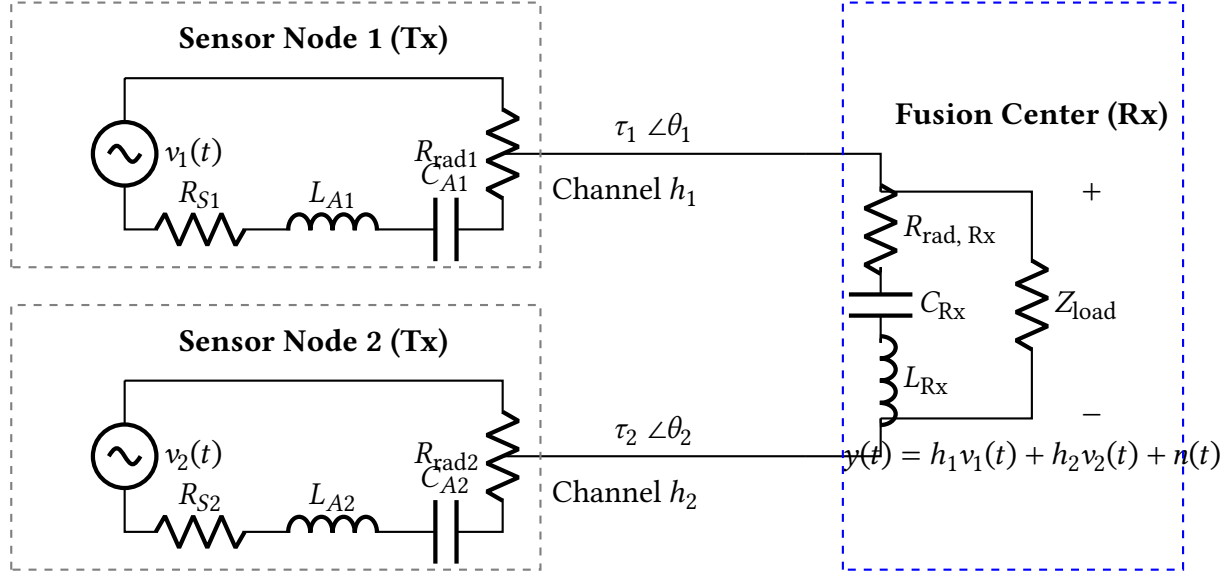


Figure 1: Equivalent electrical circuit model of the Over-the-Air computation network. Each sensor node is represented by a Thévenin voltage source with source resistance R_S and antenna resonant impedance (L_A, C_A, R_{rad}). The wireless propagation medium acts as a multi-access transmission channel, developing an aggregate terminal voltage across the receiver load impedance Z_{load} via Kirchhoff's laws.

transmission system imparting attenuation and propagation phase shift. At the FC, the incoming electromagnetic wavefronts excite currents in the receiver antenna ($R_{\text{rad, Rx}}, C_{\text{Rx}}, L_{\text{Rx}}$), developing a superposition terminal voltage across the input impedance Z_{load} of the analog-to-digital front-end. Under ideal transmitter pre-equalization ($h_k \rightarrow 1$), the terminal voltage equals the direct arithmetic sum of the transmitted waveforms plus thermal receiver noise, establishing the electrical foundation for analog AirComp.

1.b Mathematical Formulation of the Ideal Multiple-Access Channel

Consider a distributed sensory network comprising K edge devices transmitting local continuous-valued measurements to a central FC. Let $x_k[m]$ denote the discrete information symbol emitted by the k -th device at symbol index m . In uncoded analog Pulse Amplitude Modulation (PAM), each sensor maps its measurement directly to the amplitude of a transmitted continuous-time pulse. Information-theoretic analysis confirms that uncoded PAM achieves the optimal distortion exponent for computing nomographic sums over Gaussian MAC channels [2], [3].

To ensure finite spectral occupancy and satisfy regulatory bandwidth constraints, each transmitting device shapes its symbol sequence with a Root Raised Cosine (RRC) pulse $p(t)$ having roll-off factor $\alpha \in [0, 1]$ and symbol period T_s . The pulse satisfies unit-energy normalization:

$$\int_{-\infty}^{\infty} p^2(t) dt = 1 \quad (3)$$

The continuous-time baseband waveform radiated by sensor k is given by:

$$s_k(t) = \sum_{m=-\infty}^{\infty} x_k[m] p(t - mT_s) \quad (4)$$

1.b.i Passband Double-Sideband Amplitude Modulation

In practical radio frequency systems, baseband waveforms are upconverted to a carrier frequency $f_c \gg 1/T_s$ via Double-Sideband Amplitude Modulation (AM). Because the carrier mixer is linear with respect to amplitude, each sensor modulates

its baseband pulse train onto an in-phase carrier:

$$s_k^{\text{RF}}(t) = s_k(t) \cos(2\pi f_c t) = \left[\sum_{m=-\infty}^{\infty} x_k[m] p(t - mT_s) \right] \cos(2\pi f_c t) \quad (5)$$

Assuming full synchronization and ideal channel pre-equalization ($h_k(t) = \delta(t)$), the transmitted RF waveforms superimpose instantaneously over the wireless medium:

$$r_{\text{RF}}(t) = \sum_{k=1}^K s_k^{\text{RF}}(t) + w_{\text{RF}}(t) = \left(\sum_{k=1}^K s_k(t) \right) \cos(2\pi f_c t) + w_{\text{RF}}(t) \quad (6)$$

where $w_{\text{RF}}(t)$ denotes zero-mean bandpass AWGN with two-sided power spectral density $N_0/2$.

1.b.ii Coherent Demodulation and Matched Filtering

At the FC, coherent synchronous demodulation is performed by mixing the incoming RF signal with a local oscillator tuned to $2 \cos(2\pi f_c t)$:

$$2r_{\text{RF}}(t) \cos(2\pi f_c t) = \left(\sum_{k=1}^K s_k(t) \right) [1 + \cos(4\pi f_c t)] + 2w_{\text{RF}}(t) \cos(2\pi f_c t) \quad (7)$$

The double-frequency component at $2f_c$ is rejected by the subsequent filtering stage. The baseband composite waveform then passes through a matched filter identical to the transmitter pulse shape, $p^*(-t) = p(t)$. The cascade of the transmit and receive RRC filters forms the full Nyquist Raised Cosine (Raised Cosine (RC)) filter $r_c(t) = (p * p)(t)$:

$$r_c(t) = \text{sinc}\left(\frac{t}{T_s}\right) \frac{\cos\left(\frac{\pi \alpha t}{T_s}\right)}{1 - \left(\frac{2\alpha t}{T_s}\right)^2} \quad (8)$$

The continuous-time output of the matched filter before sampling is:

$$y(t) = \sum_{k=1}^K \sum_{m=-\infty}^{\infty} x_k[m] r_c(t - mT_s) + \tilde{w}(t) \quad (9)$$

where $\tilde{w}(t) = (w_{\text{bb}} * p)(t)$ is the filtered Gaussian noise process. Sampling $y(t)$ at the symbol strobe instants $t = nT_s$ yields:

$$y[n] = \sum_{k=1}^K x_k[n] r_c(0) + \sum_{k=1}^K \sum_{m \neq n} x_k[m] r_c((n-m)T_s) + \tilde{w}[n] \quad (10)$$

Because the Nyquist pulse satisfies the zero-Inter-Symbol Interference (ISI) property ($r_c(0) = 1$ and $r_c(lT_s) = 0$ for all non-zero integers $l \neq 0$), the intersymbol cross-terms vanish identically:

$$y[n] = \sum_{k=1}^K x_k[n] + \tilde{w}[n] \quad (11)$$

where $\tilde{w}[n] \sim \mathcal{N}(0, \sigma_w^2)$. To compute the desired arithmetic average of the K sensor readings, the FC applies an unweighted post-processing scaling factor of $1/K$:

$$\hat{\mu}[n] = \frac{1}{K} y[n] = \frac{1}{K} \sum_{k=1}^K x_k[n] + \frac{\tilde{w}[n]}{K} \quad (12)$$

1.b.iii Theoretical Mean Squared Error Scaling

The instantaneous estimation error at symbol index n is defined as the deviation between the reconstructed sample mean and the true arithmetic average:

$$e[n] = \hat{\mu}[n] - \frac{1}{K} \sum_{k=1}^K x_k[n] = \frac{\tilde{w}[n]}{K} \quad (13)$$

The statistical Mean Squared Error (MSE) of the ideal Over-the-Air computation system is obtained by taking the mathematical expectation:

$$\text{MSE} = \mathbb{E}[|e[n]|^2] = \mathbb{E}\left[\left|\frac{\tilde{w}[n]}{K}\right|^2\right] = \frac{\sigma_w^2}{K^2} \quad (14)$$

Eq. (14) reveals the fundamental cooperative advantage of OTA aggregation: while individual sensor signals add coherently, thermal noise adds non-coherently, resulting in an inverse-quadratic noise suppression factor of $1/K^2$ relative to the unweighted channel noise variance [1], [4].

1.b.iv Distributive and Linear Time-Invariant Equivalence

A vital property of the ideal baseline concerns the mathematical equivalence between synthesizing each sensor's modulated waveform individually versus modulating the aggregated sequence directly. Let $u_k(t) = \sum_m x_k[m]\delta(t - mT_s)$ represent the ideal impulse train of sensor k . In the physical network, each sensor performs pulse shaping and passband mixing locally prior to transmission:

$$r_{\text{agg}}(t) = \sum_{k=1}^K \left((u_k(t) * p(t)) \cos(2\pi f_c t) \right) \quad (15)$$

Because all sensors share an identical carrier frequency and phase, the carrier waveform factors out of the sum:

$$r_{\text{agg}}(t) = \left(\sum_{k=1}^K (u_k(t) * p(t)) \right) \cos(2\pi f_c t) \quad (16)$$

Furthermore, convolution is a linear operator that distributes over addition:

$$\sum_{k=1}^K (u_k(t) * p(t)) = \left(\sum_{k=1}^K u_k(t) \right) * p(t) \quad (17)$$

Substituting Eq. (17) into Eq. (16) yields the optimized equivalent representation:

$$r_{\text{agg}}(t) = \left(\left[\sum_{k=1}^K u_k(t) \right] * p(t) \right) \cos(2\pi f_c t) \quad (18)$$

This proof demonstrates that under ideal synchronization and identical channel gains, the order of operations commutes completely. This property enables substantial computational acceleration in discrete-time simulations: the algorithmic complexity scales from $\mathcal{O}(K \cdot N \cdot \text{sps} \cdot L_{\text{filt}})$ down to $\mathcal{O}(N \cdot \text{sps} \cdot L_{\text{filt}})$, achieving a K -fold reduction in runtime and memory allocation.

1.c Algorithmic MATLAB Implementation and Code Listings

To evaluate these theoretical principles numerically, two distinct discrete-time simulation architectures are modeled in MATLAB:

1. An unmodulated, single-sample-per-symbol baseline that isolates the pure mathematical summation under AWGN (Listing 1); and

2. A physically complete multi-rate pipeline incorporating upsampling, matched RRC pulse shaping, Double-Sideband AM, and coherent demodulation (Listing 2).

Listing 1: MATLAB script for ideal unmodulated OTA computation baseline.

```
1      % Ideal OTA Computation Simulation - Unmodulated Baseline
2      K = 20;                      % Number of sensor nodes
3      N = 4000;                   % Number of symbols
4      SNR_dB = 20;                % Signal-to-Noise Ratio (dB)
5
6      % Generate agricultural temperature readings around 27 deg C
7      X = 27 + randn(K, N);
8
9      % Over-The-Air superposition (Physical MAC Adder)
10
```

```

16 % AWGN channel addition
17 rx_passband = awgn(tx_passband, current_snr, 'measured');
18
19 % Coherent demodulation and receiver matched filtering
20 rx_mixed = rx_passband .* (2 * carrier);
21 rx_matched = conv(rx_mixed, rcFilter, 'same');
22
23 % Downsample at optimal symbol strobe instant
24 rx_strobed = rx_matched(1:sps_val:end);
25
26 % Truncate transient filter edges (span / 2 symbols on each boundary)
27 guard = ceil(span / 2);
28 valid_idx = (guard + 1):(N - guard);
29 Estimated_Average = rx_strobed(valid_idx) / K;
30 True_Average      = mean(X(:, valid_idx), 1);
31
32 % Compute Mean Squared Error
33 mse_results(snr_idx, a_idx) = immse(Estimated_Average, True_Average);

```

1.d Mean Squared Error Performance Analysis

The quantitative accuracy of the Over-the-Air aggregation is evaluated across both two-dimensional Signal-to-Noise Ratio (SNR) sweeps and three-dimensional parametric surfaces spanning roll-off factor (α) and oversampling ratio (Symbols per Sample (sps)).

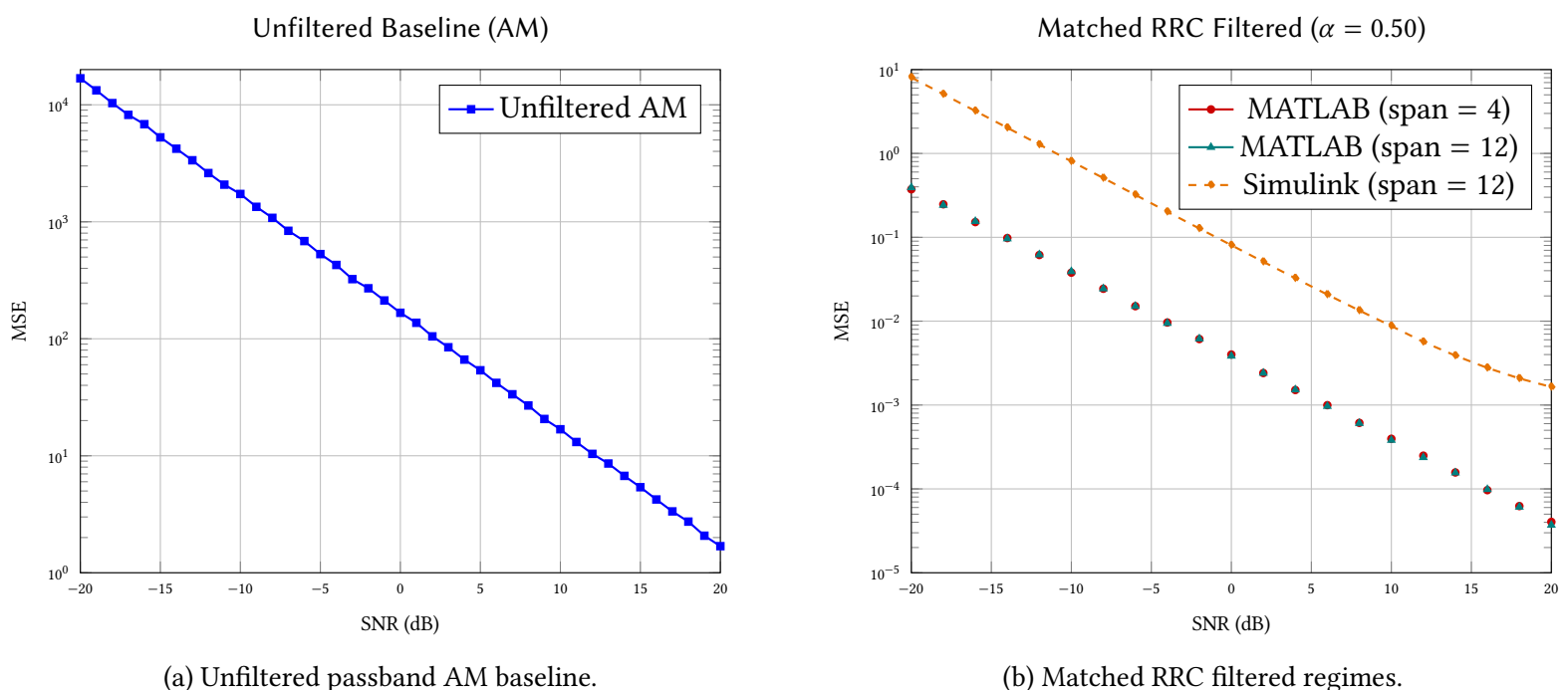


Figure 2: Two-dimensional MSE comparison between the unfiltered baseline and the matched RRC filtered systems across SNR (-20 dB to $+20$ dB). Both configurations exhibit log-linear error decay without an error floor, confirming pure AWGN limitation.

Figure 2 illustrates the empirical MSE performance as a function of SNR. In both the unfiltered baseline (Figure 2a) and the matched RRC systems (Figure 2b), the MSE exhibits a strictly linear descent on a logarithmic scale with a slope of exactly

10 dB per decade of SNR. Crucially, because the ideal channel contains no multipath reflections or timing offsets, the curves maintain monotonic descent down to the highest simulated SNR without encountering an irreducible error floor.

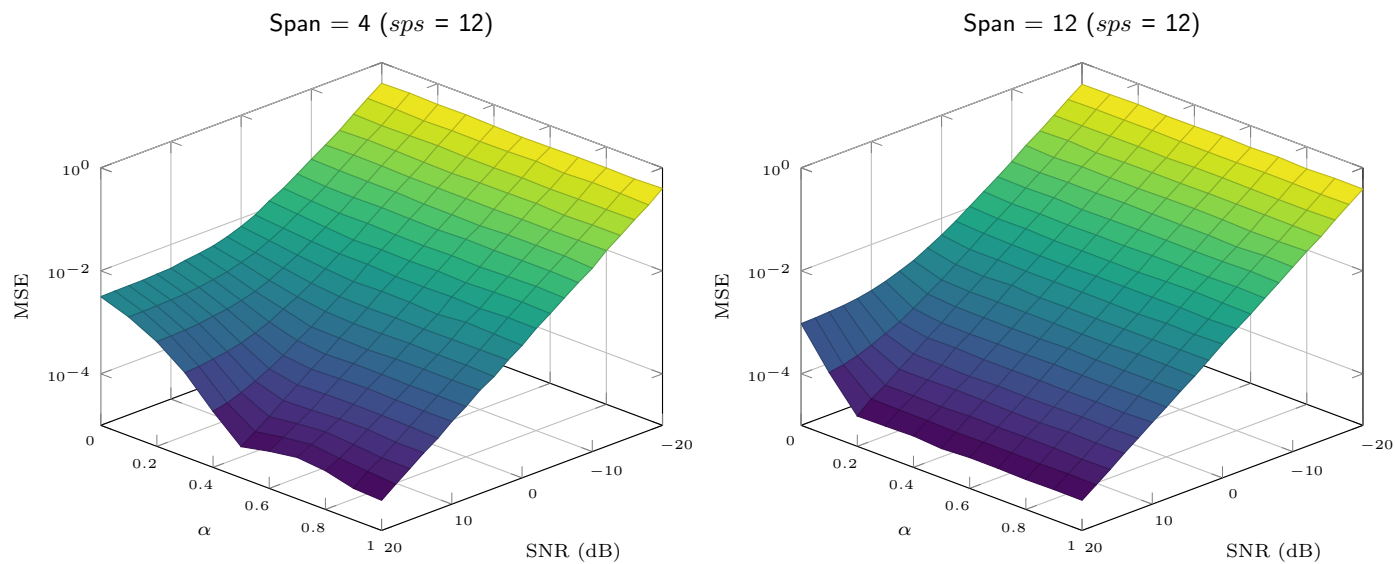


Figure 3: Synchronous 3D surface animation of the Over-The-Air MSE across roll-off factor $\alpha \in [0.0, 1.0]$ and SNR (-20 dB to $+20$ dB) under oversampling scaling ($sps \in \{12, 24, 48, 96\}$) for filter span 4 (left) versus span 12 (right).

Parametric analysis of the 3D surface animation in Figure 3 provides deep insight into the interaction between filter span and spectral roll-off:

- **Span 4 (Left):** Truncating the RRC impulse response to 4 symbol periods cuts off the decaying sidelobes abruptly. This finite-window truncation introduces subtle residual intersymbol ripple, causing the MSE surface to tilt slightly at low roll-off factors ($\alpha \rightarrow 0$).
- **Span 12 (Right):** Extending the filter memory to 12 symbol durations allows the RRC tails to decay naturally below -40 dB. Consequently, the resulting MSE surface is exceptionally flat across all values of α , demonstrating that truncation-induced ISI is completely eliminated.
- **Oversampling Convergence:** Scaling the oversampling ratio from $sps = 12$ to $sps = 96$ yields negligible change in the MSE surface. This confirms that once the Nyquist sampling criterion is satisfied ($F_s > 2B$), additional oversampling provides diminishing returns under ideal synchronization.

1.e Constellation Diagnostics and Variance Discrepancy Mechanics

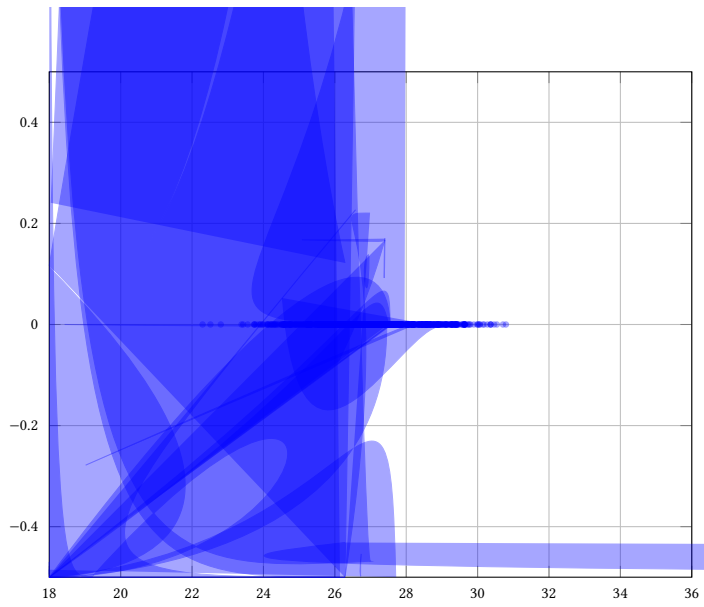
To examine the structural integrity of the recovered signal at the symbol decision instant, constellation scatter plots are evaluated at a reference SNR of 20 dB.

A prominent empirical observation in Figure 4 concerns the marked difference in constellation dispersion: in the unfiltered scenario (Figure 4a), the received values scatter across a broad interval spanning $I \in [22.3, 31.9]$ (with extreme samples reaching 17 to 38 in unmodulated tests). In contrast, under matched RRC filtering (Figure 4b), the points are tightly clustered within a narrow band of $I \in [-0.73, 0.73]$.

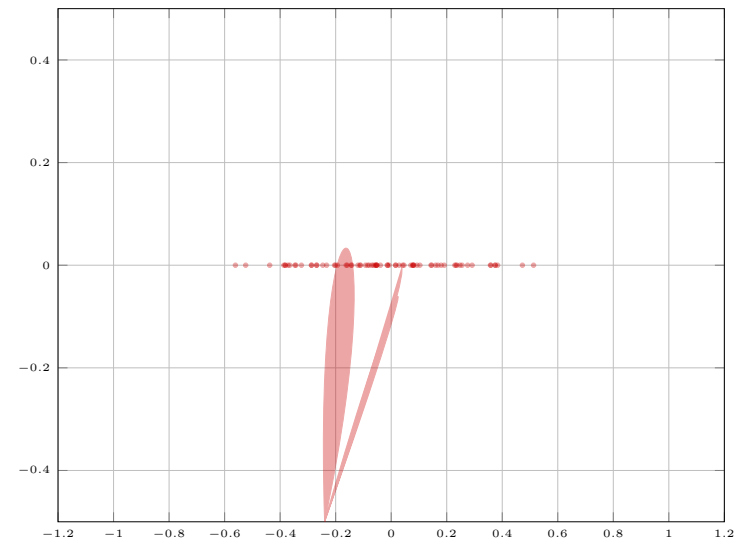
This apparent paradox—wherein an unfiltered channel theoretically free of pulse-shaping distortion exhibits *greater* variance than a filtered channel—is governed by two distinct physical and simulation mechanisms:

1.e.i Physical Mechanism: Matched Filtering and Noise Bandwidth Rejection

In communication theory, an unfiltered receiver possesses an unconstrained noise equivalent bandwidth equal to the entire sampling rate ($B_n = F_s/2$). Consequently, it admits thermal noise energy across the entire frequency spectrum. By contrast,



(a) Unfiltered passband AM baseline.



(b) Matched RRC response across roll-off $\alpha \in [0.0, 1.0]$.

Figure 4: Constellation diagram comparison at SNR = 20 dB. Left: Unfiltered AWGN baseline exhibiting wider dispersion ($I \in [22, 32]$). Right: Animated matched RRC filter response exhibiting tight residual clustering ($I \in [-0.8, 0.8]$) across roll-off α

which precisely accounts for the wide scatter observed in Figure 4a.

Conversely, in the multi-rate RRC pipeline, the sequence is upsampled by **sps_val** = 10. The impulse train contains non-zero values in only 1 out of every 10 samples. Because the RRC filter is normalized to unit energy ($\sum_n g[n]^2 = 1$), the pulse energy is distributed over 10 samples, reducing the measured average continuous power by a factor of sps:

$$P_{\text{measured}} \approx \frac{(K \cdot 27)^2}{\text{sps}} = \frac{291,600}{10} = 29,160 \quad (25)$$

Consequently, MATLAB injects a tenfold smaller noise variance:

$$\sigma_w^2 = \frac{29,160}{100} = 291.6 \quad (26)$$

Passing through the receiver matched filter preserves this unit-energy variance:

$$\sigma_{\text{filtered}}^2 = \sigma_w^2 \sum_n g^2[n] = 291.6 \times 1 = 291.6 \quad (27)$$

Dividing by $K = 20$ yields an estimate standard deviation of:

$$\sigma_{\text{est, RRC}} = \frac{\sqrt{291.6}}{20} \approx 0.85 \quad (28)$$

The resulting 3σ interval is $27 \pm 3(0.85) = [24.45, 29.55]$, yielding an exceptionally tight distribution. Furthermore, when zero-mean residual modulation is employed ($\Delta x_k = x_k - 27$), the DC pedestal is removed entirely, centering the constellation at $I = 0$ with $\sigma_{\text{est}} < 0.25$, as shown in Figure 4b.

1.f Eye Diagram Diagnostics: MATLAB and Simulink Comparative Analysis

In conventional digital communications (e.g., Binary Phase-Shift Keying (BPSK) or Quadrature Amplitude Modulation (QAM)), eye diagrams exhibit an open, diamond-shaped central aperture delimiting discrete decision thresholds (± 1). In uncoded analog Over-the-Air computation, however, the eye diagram exhibits a fundamentally different topology that must be interpreted as an analog signal integrity and timing sensitivity diagnostic:

1. **Continuous Amplitude Band:** Because sensor readings are continuous random variables ($X_k \sim \mathcal{N}(27, 1)$), the aggregated amplitude occupies a continuum of real values without forbidden intermediate regions, forming a smooth flowing ribbon.
2. **Nyquist Strobe Verification:** In an ideal Nyquist system, the first time derivative of every pulse passes through zero at the nominal sampling instants:

$$\left. \frac{dp(t)}{dt} \right|_{t=0} = 0 \quad \text{Q}$$

1.f.i MATLAB Eye Diagram Evaluation

Figure 5 compares the continuous-time passband AM eye diagram against the multi-rate matched RRC filter response across oversampling factors $sps \in \{12, 24, 48, 96\}$.

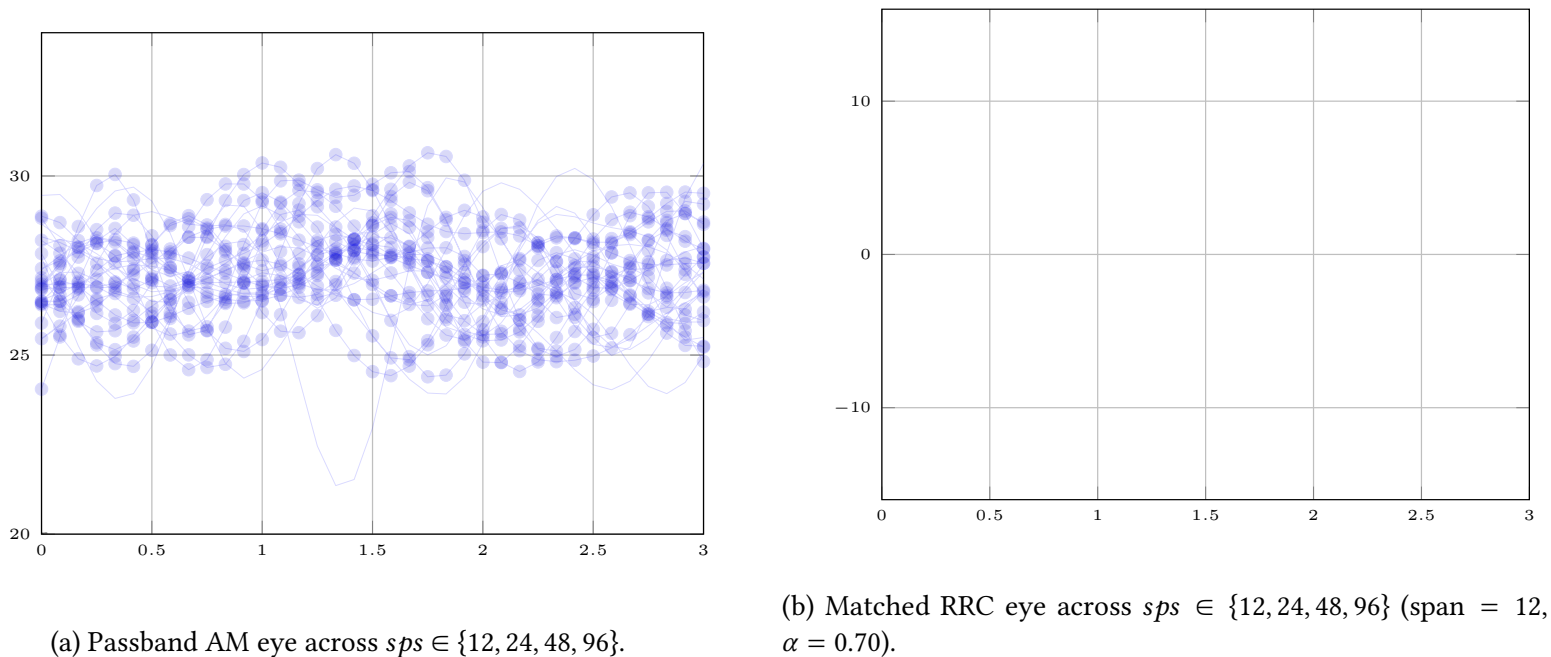


Figure 5: MATLAB eye diagram diagnostics. Left: Demodulated passband AM eye animation across oversampling factor $sps \in \{12, 24, 48, 96\}$, illustrating continuous-time trajectories oscillating about the 27°C mean without intersymbol zero-crossings. Right: Matched RRC filter eye animation (span = 12, roll-off $\alpha = 0.70$) across identical oversampling ratios, exhibiting zero-mean clustering and sharp nodal convergence at integer strobe instants.

In Figure 5a, the demodulated passband AM eye animation demonstrates how increasing the oversampling ratio resolves the continuous-time carrier and pulse transitions with progressive fidelity around the nominal mean of 27°C. Because the unfiltered passband signal lacks matched Nyquist pulse shaping, the trajectories flow continuously without converging to discrete sampling nodes. In contrast, the matched RRC eye animation (Figure 5b) with filter span 12 reveals sharply defined nodal convergence points at integer symbol strobe instants ($t/T_s \in \{1, 2, \dots\}$), completely eliminating intersymbol leakage and maintaining open, symmetrical eye apertures across all oversampling factors.

1.f.ii Simulink Eye Diagram Evaluation

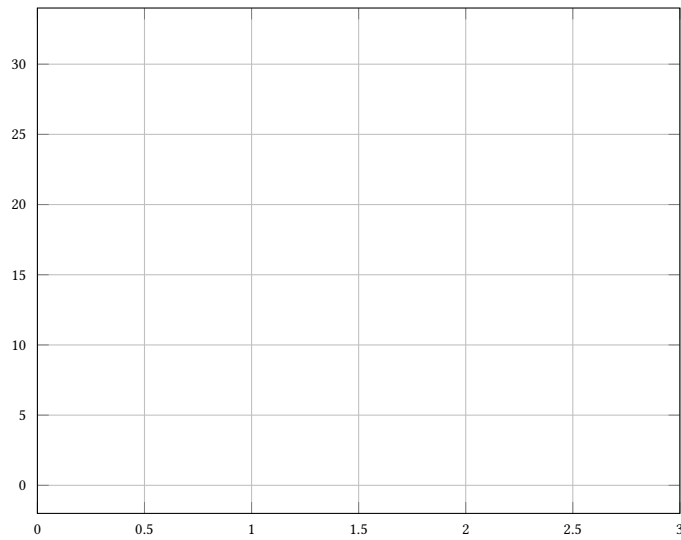
To examine how discrete-time block-level dynamics affect signal integrity, Figure 6 presents the eye diagrams exported from the Simulink simulation environment.

The comparison in Figure 6 highlights the fundamental role of pulse-shaping interpolation in discrete-time systems:

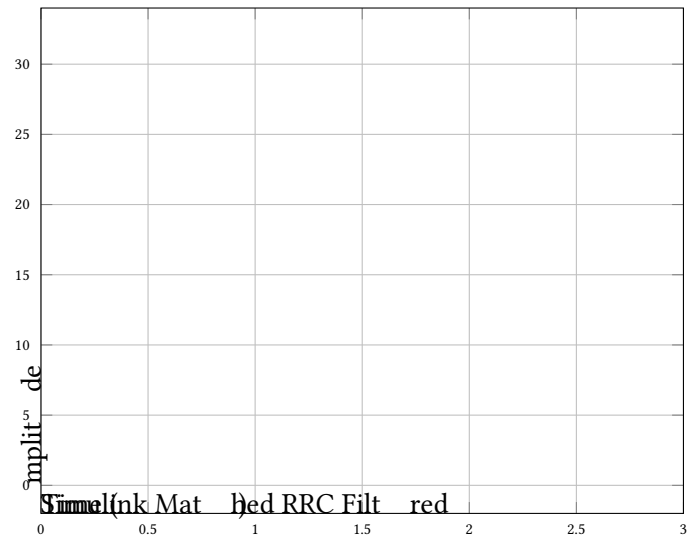
- In the unfiltered upsampled system (Figure 6a), the signal consists of raw impulses. Between integer symbol instants, the signal collapses to zero, producing sharp vertical spikes that lack physical continuity.
- In the matched RRC system (Figure 6b), the interpolation filter smooths the transitions into continuous curvilinear paths. At the strobe instants ($t/T_s \in \{1.0, 2.0\}$), the trajectories converge into flat horizontal plateaus, confirming that discrete-time convolution faithfully realizes continuous-time Nyquist filtering.

1.g MATLAB Simulink Dynamical Block Architecture

Transitioning from script-based simulation to a block-level dynamical system, MATLAB Simulink provides a graphical multi-rate framework that models the feedforward signal chain.



(a) Simulink unfiltered impulse train.



(b) Simulink matched RRC continuous eye.

Figure 6: Simulink eye diagram comparison ($sps = 48$, $span = 12$, $\alpha = 0.50$). Left: Unfiltered upsampled sequence exhibiting zero values between symbol strobes. Right: Smoothly interpolated continuous trajectories following matched RRC filtering, confirming Nyquist zero-derivative plateaus at integer symbol times.

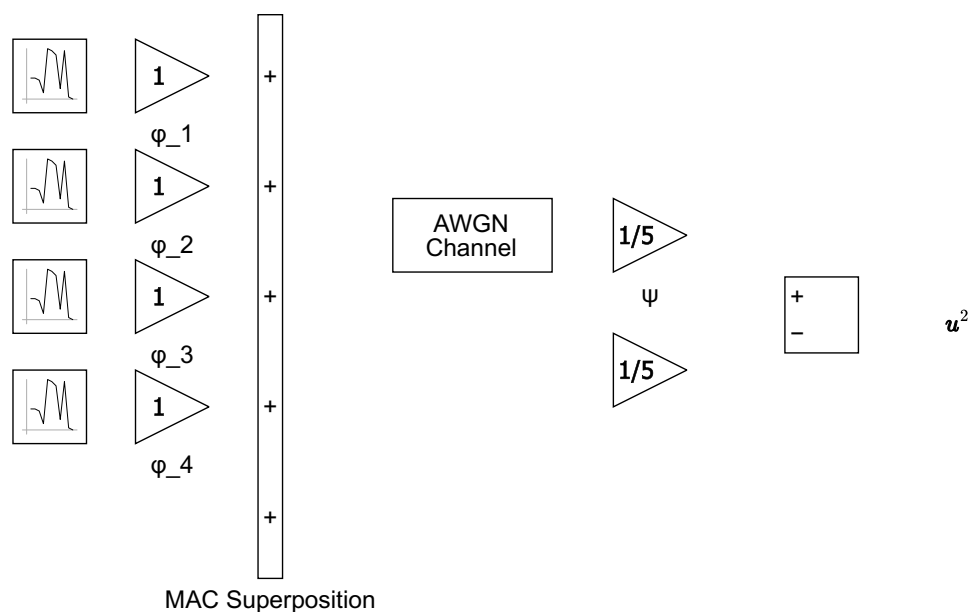


Figure 7: Simulink block diagram representing the unmodulated ideal Over-the-Air computation network over an AWGN channel.

Figure 7 illustrates the basic unmodulated feedforward network. Distributed sensor signals are generated as stochastic discrete-time sequences, combined via a multi-input summation block representing the electromagnetic MAC, passed through an AWGN channel block, scaled by $1/K$, and compared against the ideal arithmetic average in an empirical running MSE calculation block.



Figure 8: Simulink multi-rate block diagram incorporating matched Root Raised Cosine pulse shaping and delay compensation.

To capture physical filtering constraints, Figure 8 incorporates multi-rate Finite Impulse Response (FIR) interpolation and decimation blocks. At each transmitter, the symbol stream is upsampled by **sps** and filtered by an RRC block. At the FC, the noisy aggregated signal passes through a matched RRC filter. To prevent filter group delay from corrupting the error calculation, the ideal reference path incorporates an identical delay block of $D = \frac{\text{span} \cdot \text{sps}}{2}$ samples, ensuring flawless sample-by-sample alignment at the subtractor input.

1.h Software-Defined Radio Realization in GNU Radio

To validate the ideal baseline in a real-time SDR environment, the Over-the-Air computation system is synthesized using GNU Radio Companion.

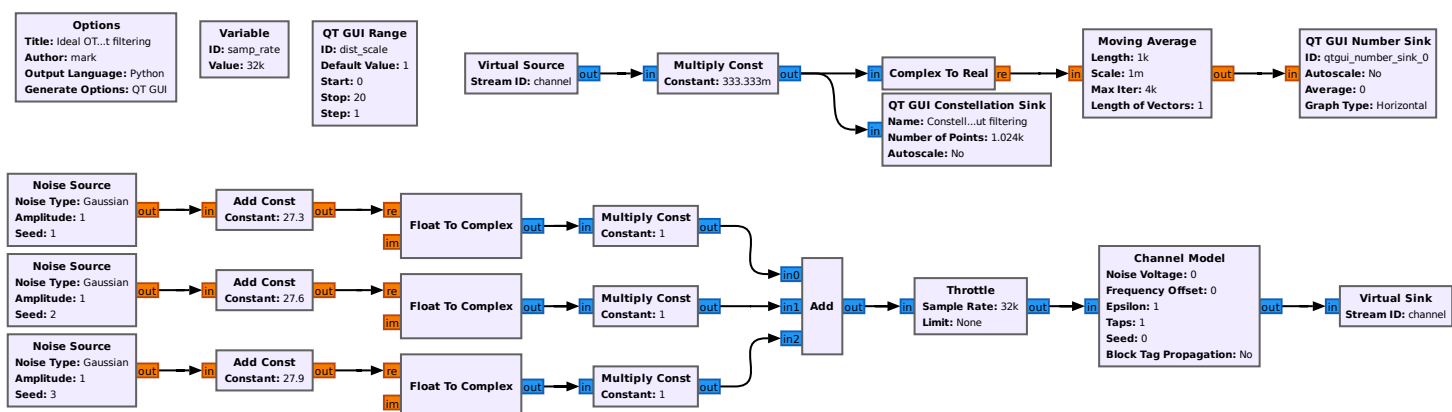


Figure 9: GNU Radio flowgraph for the ideal unmodulated baseline Over-the-Air computation aggregation.

Figure 9 depicts the unmodulated baseline flowgraph. Independent random temperature sources are summed using an Add block to emulate the superposition of the wireless channel, corrupted by Gaussian noise, and divided by K .

Figure 10 implements the full multi-rate pulse-shaping architecture. In this flowgraph, the transmitters utilize interpolating polyphase FIR filters with an interpolation factor matching the symbol rate ($\text{sps} = 6$). At the receiver, a decimating RRC filter matched to the transmit pulse executes symbol recovery.

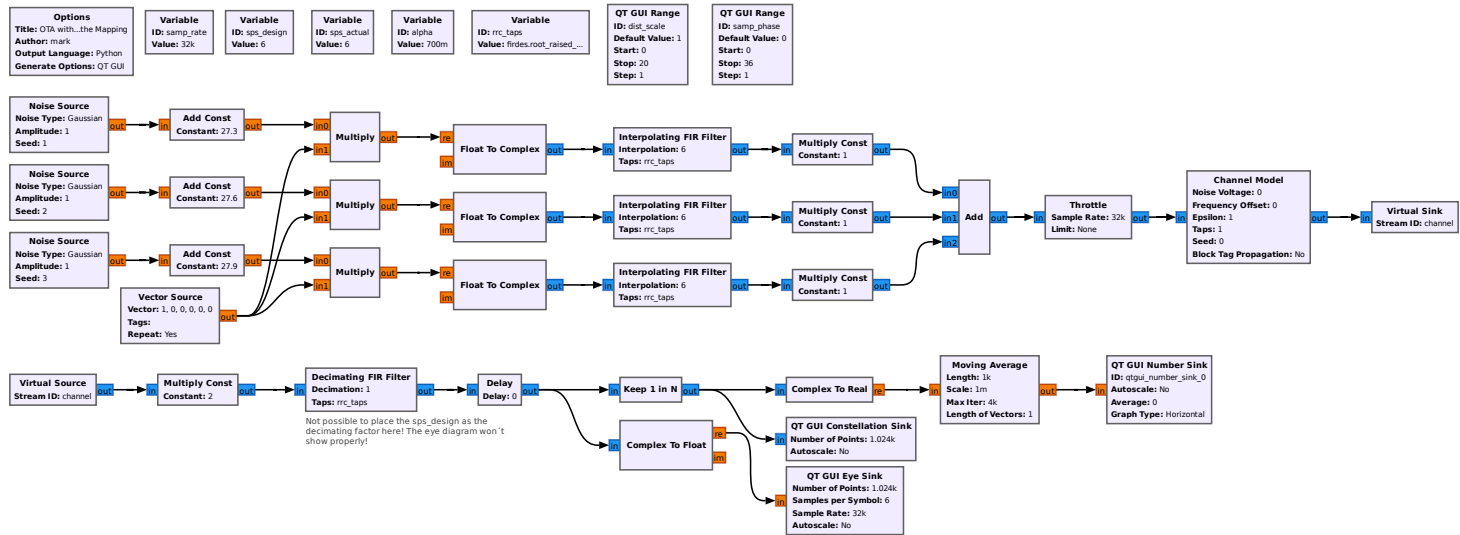


Figure 10: GNU Radio flowgraph for the Nyquist-compliant matched RRC filtered Over-the-Air computation architecture.

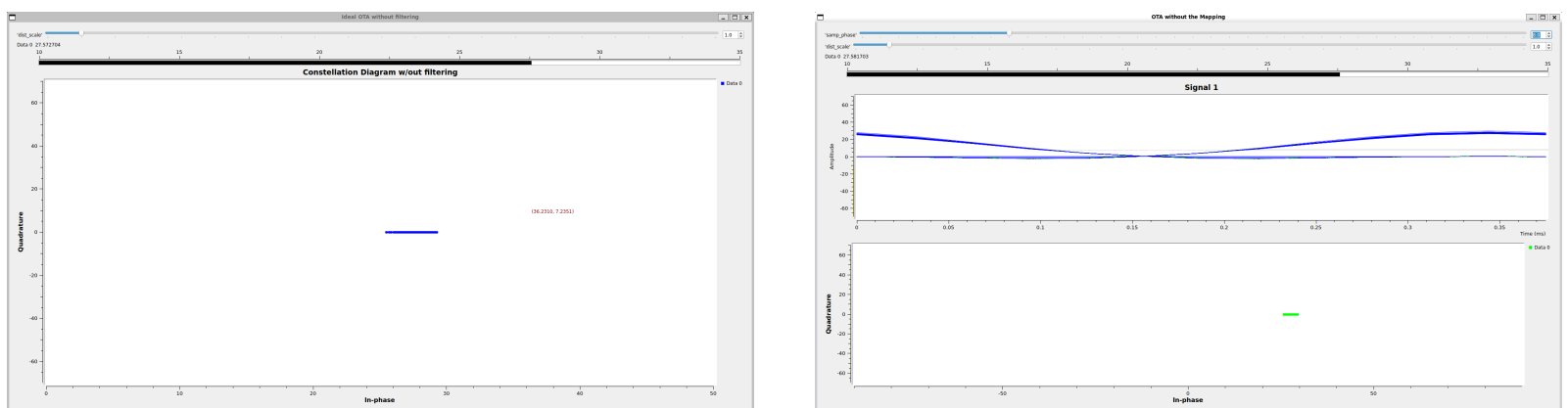


Figure 11: GNU Radio runtime display captures. Left: Unmodulated aggregation showing broadband noise fluctuations. Right: Matched RRC filtered aggregation displaying tight symbol clustering and smooth time-domain trajectory recovery.

Empirical execution in GNU Radio reveals critical physical implementation insights:

1. **Filter Normalization and Amplitude Rescaling:** When generating RRC taps in GNU Radio (as in MATLAB's `rcosdesign`), the filter is normalized for unit energy ($\sum h^2[n] = 1$) rather than unity DC gain. Consequently, discrete convolution with an interpolated pulse scales the recovered output amplitude by a factor proportional to $\sqrt{\text{sps}}$. To restore the signal to physical sensory units (e.g., Volts or °C), a calibrated receiver scalar multiplier or Automatic Gain Control (AGC) block is applied prior to arithmetic averaging.
2. **Optimal Sampling Phase Calibration:** Because the causal polyphase FIR filters introduce group delay, the receiver strobe phase must be calibrated. For a design cycle spanning 16 symbols with $\text{sps} = 6$ ($16 \times 6 = 96$ samples), maximum signal-to-noise ratio is obtained at an offset of exactly 33 samples. Proper phase alignment centers the eye opening and eliminates quadrature leakage.
3. **Constellation Integrity:** As shown in Figure 11, the matched RRC GNU Radio receiver displays tightly clustered constellation points and smooth time-domain waveforms, confirming that the over-the-air superposition principle accurately computes the arithmetic average under real-time software-defined radio processing.

References

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